

# Assignment 2: Time Series Forecasting with Classical and Deep Learning Methods

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**Deadline: 2025-05-11 23:59 (UTC+8) Grade: 10%**

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## Background

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Time series forecasting is a fundamental task in time series analysis, involving the prediction of future values based on historical data. In this assignment, you will apply both a **classical forecasting method** and a **deep learning-based method** to perform multivariate time series forecasting on the `ETT` dataset.

## Requirements

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You are required to implement and evaluate two forecasting models:

- A **classical forecasting method**: Choose either **ARIMA** or **Prophet**.
- A **deep learning-based forecasting method**: Choose one from **RNN**, **Transformer**, or other deep learning models.

The forecasting task should be conducted on the following four subsets of the ETT dataset (available at: [https://drive.google.com/file/d/1bnrv7gpn27yO54WJl-vuXP5NcIE5BIBx/view?usp=drive\\_link](https://drive.google.com/file/d/1bnrv7gpn27yO54WJl-vuXP5NcIE5BIBx/view?usp=drive_link)):

- `ETTh1`
- `ETTh2`
- `ETTm1`
- `ETTm2`

## Forecasting Settings

- **For the classical method (ARIMA or Prophet):**
  - Perform **univariate forecasting**.
  - Use only the **OT** column for both training and prediction.
- **For the deep learning method (RNN, Transformer, etc.):**
  - Perform **multivariate forecasting**.
  - Use **all available features** as input to predict the **OT** column.

## Experimental Settings

- **Input window size**: 96
- **Prediction length**: 96 and 192
- **Normalization**: Standardize each feature (zero mean, unit variance) based on training data statistics. Apply the same normalization to validation and test sets.

- **Dataset splitting:** Split the data into training, validation, and test sets according to a **3:1:1** ratio in time order.

## Evaluation

Evaluate your models using the following metrics on the **normalized** data:

- **MSE (Mean Squared Error)**
- **MAE (Mean Absolute Error)**

You should report the MSE and MAE results for each method on each of the four datasets.

## Submission

You can submit your assignment in one of the following formats:

- **Option 1:** A zip containing:
  - Source code
  - A PDF file summarizing:
    - The selected methods.
    - Experimental settings.
    - MSE and MAE results.
    - A short discussion on the observed forecasting performance.
- **Option 2:** A single Jupyter Notebook (.ipynb) that:
  - Includes all code and results.
  - Uses markdown cells to explain methods, experiments, and results.
  - Summarizes the final MSE and MAE results in a clear table.
  - A short discussion on the observed forecasting performance.

**Please submit your assignment on web-learning (<https://learn.tsinghua.edu.cn>)**

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If you have any questions, please contact TA-Zhe Xie: [xiez22@mails.tsinghua.edu.cn](mailto:xiez22@mails.tsinghua.edu.cn)